

SHETH L.U.J & SIR MV COLLEGE

Aim:RULE MINING

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

from mlxtend.frequent_patterns import apriori, association_rules

# Load the Superstore dataset
df = pd.read_csv('Superstore.csv', encoding='latin1')

# Display the first five records
print("First five records:")
print(df.head())

# Create transactions using Order ID and Sub-Category
basket = pd.crosstab(
    df['Order ID'],
    df['Sub-Category']
)

# Convert the transaction values into True and False
basket = basket > 0

print("\nTransaction Data:")
print(basket.head())

# Find frequent itemsets using the Apriori algorithm
frequent_itemsets = apriori(
    basket,
    min_support=0.01,
    use_colnames=True
)

print("\nFrequent Itemsets:")
print(frequent_itemsets)

# Generate association rules using confidence
rules = association_rules(
    frequent_itemsets,
```

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```
metric='confidence',
min_threshold=0.20
)

# Sort the rules according to confidence
rules = rules.sort_values(
    by='confidence',
    ascending=False
)

# Display support, confidence and lift
print("\nAssociation Rules:")
print(
    rules[
        ['antecedents',
         'consequents',
         'support',
         'confidence',
         'lift']
    ].head(10)
)

# Create readable names for the rules
top_rules = rules.head(10).copy()

top_rules['rule'] = (
    top_rules['antecedents'].apply(
        lambda x: ', '.join(list(x))
    )
    + ' -> ' +
    top_rules['consequents'].apply(
        lambda x: ', '.join(list(x))
    )
)

# Visualize support and confidence of the top rules
plt.figure(figsize=(12, 6))

plt.bar(
    top_rules['rule'],
    top_rules['support'],
    label='Support'
)
```

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```
plt.bar(
    top_rules['rule'],
    top_rules['confidence'],
    bottom=top_rules['support'],
    label='Confidence'
)

plt.xlabel('Association Rules')
plt.ylabel('Value')
plt.title('Support and Confidence of Top Association Rules')
plt.xticks(rotation=45, ha='right')
plt.legend()

plt.tight_layout()
plt.show()

# Visualize the lift of the top rules
plt.figure(figsize=(12, 6))

plt.bar(
    top_rules['rule'],
    top_rules['lift']
)

plt.xlabel('Association Rules')
plt.ylabel('Lift')
plt.title('Lift of Top Association Rules')
plt.xticks(rotation=45, ha='right')

plt.tight_layout()
plt.show()
```

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Output:

```
C:\Windows\py.exe
Python version: 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]
Python location: C:\Python314\python.exe

First five records:
Row ID  Order ID  Order Date  Ship Date  Ship Mode  ...  Product Name  Sales  Quantity  Discount  Profit
0      1  CA-2016-152156  11/8/2016  11/11/2016  Second Class  ...  Bush Somerset Collection Bookcase  261.9600  2  0.00  41.9136
1      2  CA-2016-152156  11/8/2016  11/11/2016  Second Class  ...  Hon Deluxe Fabric Upholstered Stacking Chairs,...  731.9400  3  0.00  219.5820
2      3  CA-2016-138688  6/12/2016  6/16/2016  Second Class  ...  Self-Adhesive Address Labels for Typewriters b...  14.6200  2  0.00  6.8714
3      4  US-2015-108966  10/11/2015  10/18/2015  Standard Class  ...  Bretford CR4500 Series Slim Rectangular Table  957.5775  5  0.45  -383.0310
4      5  US-2015-108966  10/11/2015  10/18/2015  Standard Class  ...  Eldon Fold 'N Roll Cart System  22.3600  2  0.20  2.5164

[5 rows x 21 columns]

Transaction Data:
Sub-Category  Accessories  Appliances  Art  Binders  Bookcases  Chairs  Copiers  Envelopes  ...  Furnishings  Labels  Machines  Paper  Phones  Storage  Supplies  Tables
Order ID
CA-2014-100006  False  False  False  False  False  False  False  False  ...  False  False  False  False  True  False  False  F
alse
CA-2014-100090  False  False  False  True  False  False  False  False  ...  False  False  False  False  False  False  False  F
alse
CA-2014-100293  False  False  False  False  False  False  False  False  ...  False  False  False  True  False  False  False  F
alse
CA-2014-100328  False  False  False  True  False  False  False  False  ...  False  False  False  False  False  False  False  F
alse
CA-2014-100363  False  False  False  False  False  False  False  False  ...  False  False  False  True  False  False  False  F
alse

[5 rows x 17 columns]

Frequent Itemsets:
support  itemsets
0  0.143342  frozenset({Accessories})
1  0.090038  frozenset({Appliances})
2  0.145937  frozenset({Art})
3  0.262727  frozenset({Binders})
4  0.044720  frozenset({Bookcases})
...
63  0.023358  frozenset({Phones, Storage})
64  0.010381  frozenset({Tables, Phones})
65  0.010182  frozenset({Binders, Paper, Furnishings})
66  0.010781  frozenset({Phones, Binders, Paper})
67  0.010581  frozenset({Binders, Paper, Storage})

[68 rows x 2 columns]
```

```
C:\Windows\py.exe
alse

[5 rows x 17 columns]

Frequent Itemsets:
support  itemsets
0  0.143342  frozenset({Accessories})
1  0.090038  frozenset({Appliances})
2  0.145937  frozenset({Art})
3  0.262727  frozenset({Binders})
4  0.044720  frozenset({Bookcases})
...
63  0.023358  frozenset({Phones, Storage})
64  0.010381  frozenset({Tables, Phones})
65  0.010182  frozenset({Binders, Paper, Furnishings})
66  0.010781  frozenset({Phones, Binders, Paper})
67  0.010581  frozenset({Binders, Paper, Storage})

[68 rows x 2 columns]

Association Rules:
antecedents  consequents  support  confidence  lift
27  frozenset({Phones, Paper})  frozenset({Binders})  0.010781  0.308571  1.174494
29  frozenset({Paper, Storage})  frozenset({Binders})  0.010581  0.297753  1.133316
9  frozenset({Fasteners})  frozenset({Binders})  0.012378  0.288372  1.097611
2  frozenset({Appliances})  frozenset({Binders})  0.025953  0.288248  1.097140
25  frozenset({Paper, Furnishings})  frozenset({Binders})  0.010182  0.288136  1.096711
10  frozenset({Fasteners})  frozenset({Paper})  0.011779  0.274410  1.154125
26  frozenset({Phones, Binders})  frozenset({Paper})  0.010781  0.271357  1.141248
28  frozenset({Binders, Storage})  frozenset({Paper})  0.010581  0.266332  1.120114
24  frozenset({Binders, Furnishings})  frozenset({Paper})  0.010182  0.257576  1.083289
15  frozenset({Storage})  frozenset({Binders})  0.039728  0.256113  0.974826
```

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Figure 1

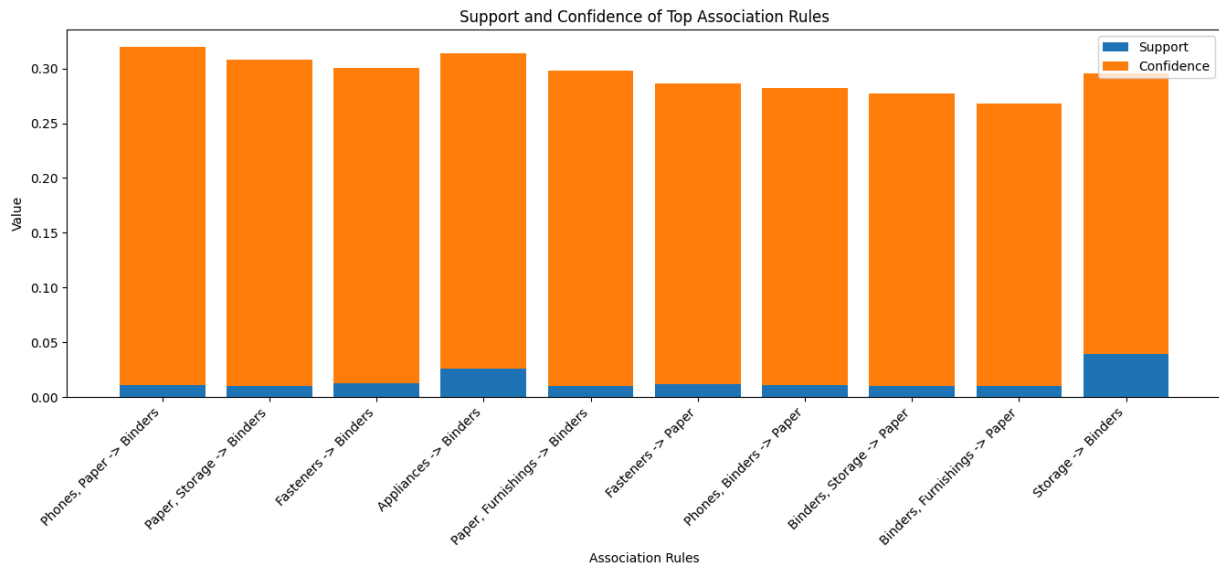


Figure 1

